Ziv Behar

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EDUCATION

University of California, Berkeley ***Expected: May '27***

B.S. Electrical Engineering and Computer Science (Honors) | *College of Engineering* GPA: 3.98/4.0

Research Advisor: Liwei Lin

Fremont High School, Sunnyvale, CA ***May '24***

RESEARCH

Liwei Lin Lab (Berkeley Sensor & Actuator Center) ***Feb '26 – Present***

Physics-Informed ML for 3D Ultrasonic Imaging ***Apr '26 – Present***

- Invented the first physics-informed learned reconstruction method for in-air 3D ultrasonic imaging, recovering target position, size, and orientation from only 8 PMUTs vs. hundreds conventionally.
- Created a differentiable acoustic forward model with learned iterative reconstruction, using fixed wave physics to recover sparse 3D surfaces from coded ultrasonic echoes.
- Eliminated mechanical scanning with an 8-element coded-array design, reducing acquisition time by 10–100× while achieving 34 mm median surface error.

PMUT Compressed-Sensing Imaging **Feb '26 – Present**

- Built compressed-sensing synthetic-aperture PMUT imaging using coded measurements and sparse reconstruction, improving spatial resolution 11.8× to 2.87 mm.
- Demonstrated the first stationary PMUT-based non-line-of-sight ultrasonic imaging method, reconstructing hidden targets up to 1.42 m away within 0.9 mm of ground truth.
- Devised a genetic-algorithm calibration method for modeling and suppressing PMUT transmit ringdown by >95%, improving recovery of weak reflected signals near the transducer.

PUBLICATIONS

1. **Z. Behar**, N. G. Lukhanin, D. Muthu, A. Shivkumar, M. Sedky, L. Lin. "Physics-Informed Learned Reconstruction for In-Air 3D Ultrasonic Imaging with a PMUT Array" IEEE Transducers, 2027. **In Preparation**
2. **Z. Behar**, N. G. Lukhanin, A. Shivkumar, M. Sedky, L. Lin. "Noncontact Material Characterization Using Piezoelectric Micromachined Ultrasonic Transducers" IEEE Transducers, 2027. **In Preparation**
3. N. G. Lukhanin, **Z. Behar**, D. Muthu, A. Shivkumar, T. Esses, M. Sedky, L. Lin. "Subwavelength Ultrasonic Imaging via PMUTs through Synthetic Aperture" IEEE MEMS, 2027.
4. N. G. Lukhanin, D. Muthu, **Z. Behar**, A. Shivkumar, T. Esses, M. Sedky, L. Lin. "Non-Line-of-Sight 3D Imaging Using Piezoelectric Micromachined Ultrasound Transducers" IEEE MEMS, 2027.
5. N. G. Lukhanin, **Z. Behar**, D. Muthu, A. Shivkumar, T. Esses, M. Sedky, L. Lin. "Compressive 3D Imaging Using Co-Optimized Piezoelectric Micromachined Ultrasound Transducer Array" 2026. **In Preparation**

TALKS & PRESENTATIONS

Gulf Coast Undergraduate Research Symposium (GCURS) **Oct '26**

- Selected from a highly competitive international applicant pool to deliver an oral presentation on original research in the ECE division at Rice University and awarded a merit scholarship.

WORK EXPERIENCE

AWS – *Software Development Engineer Intern* ***May '26 - Aug '26***

- Migrated relational databases throttling at 1.5B+ reqs/day from an in-process system to a dedicated architecture scaling from ~50 to 500+ hosts, protecting pre/post-auth traffic for 13 request dimensions.
- Built a shadow-evaluation framework and automated rule-translation tooling to validate the new system against real-time production behavior and detect rule and decision drift.
- Optimized throttling handler and cache performance while building fail-safe behavior, production monitoring, and AI-agent-assisted CI/CD safeguards for safe rule deployment.

LIFTR *App Store* – *Software Engineer* ***Nov '25 - May '26***

- Built an ML system for an acquired, revenue-generating iOS fitness app, processing 17 pose signals from live camera input on-device to deliver personalized real-time form coaching.
- Solely developed the app end-to-end for 10,000+ users, spanning UI/UX, Firebase backend and auth, StoreKit payments, analytics, and iOS widgets/Live Activities.
- Created an LSTM-based workout recommendation engine that retrains incrementally as users log workouts, adapting recommendations over time.

AWARDS

CS152 Advanced Computer Architecture AWS CNN Optimization competition winner **SP '26**

- Optimized the CNN general implementation on AWS Trainium hardware by ~20x the baseline out of 160 UC Berkeley students.

Blockchain@Berkeley Hackathon 1st place winner **'26**

- Built STRATA, a real-time pipeline that turns live conflict news in the Middle East into structured events, analyzes and maps them to Polymarket prediction markets, then automatically place trades.

Dean's List - College of Engineering **'25, '26**

- Recognizes outstanding academic achievement. To be eligible, students must have a GPA in the top 10% of all College of Engineering undergraduates.

Fremont Union High School District Board of Trustees Community Service Award **SP '24**

- Recognized for completing 80+ hours of community service during senior year. Demonstrate outstanding character and desire to make a community a better place.

Fremont High School Principal's Honor Roll **SP '24**

- Awarded to students who held a GPA in the top 5% of the graduating class.

PROJECTS

SnapSort ***Jan '25 - Dec '25***

- Built a transformer-based event clustering system for large-scale personal camera rolls with 84% accuracy, including a reproducible data ingestion and labeling pipeline for unstructured photo streams.
- Built a mobile ↔ cloud ML pipeline for GPU-accelerated embedding generation and vector search using custom gRPC services and high-throughput parallelized inference.

MoffittStatus (acquired by ASUC) ***Sep '24 - Jan '25***

- Launched a real-time library occupancy platform showing students where open seats are available within multiple UC Berkeley libraries, reaching 3.5K users in month one and 200k+ student views.
- Built the full stack from WiFi traffic sensor ingestion and processing to occupancy estimation, REST APIs, database infrastructure, and front-end.

OrderCubic ***Oct '22 - Jun '24***

- A SaaS application designed to help companies reduce office costs by managing shared cubicles and seating plans. Include canvas editors for layouts with instant, multiuser updates using MySQL database.
- Implemented a multi-tenant reservation backend with role-based admin controls, booking conflict detection, bulk employee onboarding, two-factor authentication, and automated deployment on AWS.

GAME DEVELOPMENT (Pre AI)

Balloons Pop

- Strategy defense game where players place units to survive increasingly difficult balloon waves.
- Built data-oriented game logic for units, targeting, upgrades, enemy behavior, and wave progression.

Sticky Fred

- Wall-sticking coin-collection game where players jump through levels and dodge blades.
- Built custom physics and a level editor for objects, hazards, and particle effects.

Money Portal

- Portal-and-gravity puzzle game where players navigate linked spaces using momentum.
- Built custom physics for gravity, collisions, and movement through connected portals.

Please see zivbehar.vercel.app for information on the rest of my awesome projects.

VOLUNTEERING

Hanhagat Tzabar - Friends of the Israel Scouts, Inc., Tzofim

- Volunteered 800+ hours serving as President and Lead Counselor for a 400+ member Israeli Scouts chapter, organizing events, weekly sessions, meetings, and camps across the organization.
- Counseled 80+ high-school Scout leaders weekly and helped shape chapter-wide initiatives and decision-making across U.S. chapters.

SKILLS

Languages: Python, Java, JavaScript/TypeScript, C, C++, RISC-V, Swift, SQL.

Frameworks: PyTorch, CoreML, React/React Native, AWS, gRPC, Playwright, Logism, LTspice, KiCad.

Hardware: PCBs, KiCAD, AWG, Oscilloscopes, 3D printing, Arduino, Jetson Xavier NX.